

Sea Turtle Segmentation using U-Net and DeepLabv3+

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I. INTRODUCTION

Segmentation has been utilised across various domains and industries, serving as a crucial tool for fundamental tasks within computer vision and machine learning. The focal point of this project involves the segmentation of sea turtles, employing two deep learning methods.

SeaTurtleID2022 represents an exhaustive imagery dataset from the Sea Turtle Conservation program[18]. This dataset provides 8,729 photographs representing over 13 years of encountering turtles. Each image of this dataset is furnished with detailed segmentations of the head, flippers, and carapace parts of the turtles in question, which are essential for training all sorts of relevant segmentation models.

With the rise in deep learning, the limitations of traditional methods were greatly offset by the quick performance and accuracy provided by convolutional neural networks (CNN). Our project works with U-Net and DeepLabv3+, with modified adjustments to enhance precision for turtle segmentation. U-Net's encoder-decoder architecture is combined with MobilenetV2, to enhance novelty and assess the application of state-of-the-art principles from the biomedical domain to a different context namely turtle segmentation. The DeepLabv3+ is modified with an adapted Resnet backbone to precisely extract minute features like flippers.

Overall, the objective of our project is to create a model that can highlight and segment different parts of a sea turtle in an image, and secondly, improve the accuracy of our model.

II. LITERATURE REVIEW

A. Background of Image Segmentation

Image segmentation is one of the basic problems within computer vision. It has crucial applications in the medical

imaging, self-driving, remote sensing, and wildlife conservation applications. The goal of image segmentation aims at categorising every pixel in an image into one of several classes, effectively partitioning the image into marked regions for further analysis. In the context of the project, it involves extracting the head, flippers, and carapace from each image, which can feature noisy or cluttered backgrounds. With such a complicated task, powerful, modern segmentation methods with good generalisation across different datasets are called for.

Conventionally, before the emergence of deep learning, segmentation was performed using more traditional methods, which could not handle complex input. The following including

- **Thresholding:** Simple methods, for example, include thresholding by Otsu [2], which was applied to images of high contrast where objects could easily be made out from the background. In cases, where there is poor differentiation between the background and the object of interest, thresholding can lead to incorrect segmentation results. Leading the method to be only effective on a small subset of input images such as contrasted cell images [19]
- **Edge Detection and Watershed Transform:** Techniques that use operators like Watershed [3] have been used widely but often failed in the case of complex scenes where objects overlapped or with variable illumination conditions.

The robustness of traditional segmentation methods was offset by the resurgence of deep learning in the field. With the rise of deep learning, Convolutional Neural Networks (CNN) became a key milestone for image segmentation tasks. CNNs leverage a large quantity of labelled data for learning rich feature representations that help recognise complex patterns. Introducing Fully Convolutional Networks (FCNs) became an important milestone in semantic segmentation. FCNs replaced the fully connected layers in conventional CNNs with convolutional layers to generate dense pixel-wise

predictions [8]. Despite their success, FCN faced limitations, particularly in retaining spatial accuracy due to the pooling operations. Hence, the creation of advanced architectures, including U-Net and DeepLabv3+, was needed to fill this gap.

B. U-Net

The U-Net was first proposed by Ronneberger et al. [4] for biomedical image segmentation. It addressed the major limitation of Fully Convolutional Networks (FCNs) with a fully symmetric encoder-decoder architecture using skip connections. Since its proposition, it has been successfully applied in many fields owing to good performance on small data and the preservation of spatial accuracy.

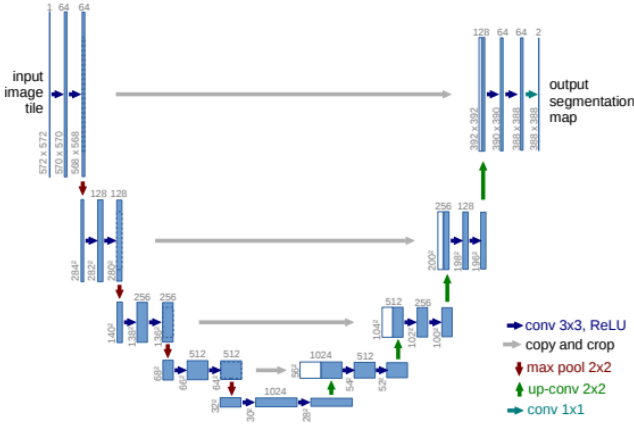


Fig 1. U-net architecture, for instance, has the lowest resolution of 32x32 pixels. Each of the blue boxes represents a multi-channel feature map. A note on top of the box indicates how many channels. The lower left border of the box shows the x-y-size. Copies of feature maps are shown in white boxes. The arrows show the various operations[4].

- **Encoder-Decoder Architecture:** The encoder down-samples the feature through a process of feature extraction. This is followed by the decoder's upsampling of the feature maps to regain the spatial resolution. Corresponding skip connections between the encoder and the decoder layers recover lost spatial information, thus allowing the network to predict accurate segmentation masks [4].
- **Application and Relevance:** U-Net has been identified to capture with ease fine details and broader contextual information which would be useful when segmenting sea turtles. Skip connections help precisely define small areas of the sea turtle anatomy, such as flippers, and the head [4].

C. MobileVnet2

MobileNet models are CNNs designed to run on mobile devices. The initial version released was MobileNetV1, proposed by Howard et al.[33], with its distinct feature being its ability to employ “depthwise separable filters” [33]. The depth-wise separable convolutions make the process computationally less expensive compared with a common convolution operation [32].

MobileNetV2 belongs to the family of MobileNets, which are faster compared to MobileNetV1 [32]. The MobileNetV2 model compared to MobileNetV1 has two more features added [32]. They are linear bottleneck layers, and secondly, shortcut connections between bottlenecks. The bottleneck layer reduces the amount of data flowing in the network. The MobileNetV2 model contains two types of bottleneck layers [32].

Considering other pre-trained models, the MobileNetV2 model has fewer parameters to train [32]. Besides that, it offers high accuracy, which requires little storage and processing power. Therefore, by replacing the encoder part of U-net model with a pre-trained MobileNetV2, the resulting architecture gives higher segmentation accuracy than the normal U-net model [32].

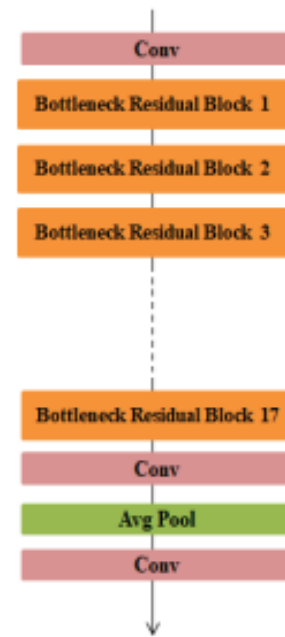


Fig 2. MobileNetV2 Architecture[32].

Advantages of using MobileNetV2 as an Encoder

As aforementioned, due to the small parameters provided by MobileNetV2, it makes it easier and faster to train. This helps the model converge much faster compared to a non-pretrained model. With such a pre-trained encoder, the model performance will be pretty high when contrasting with a non-pre-trained model. This modification has been utilised as a novel method that enhances the state of the art in the domain of medical tasks such as brain tumour segmentation, where accuracy is detrimental to the results[36].

D. DeepLabv3+

DeepLabv3+ is a segmentation model based on an encoder-decoder architecture. The encoder extracts features from the image and the decoder uses those features to determine which pixels are most likely to belong to which class.

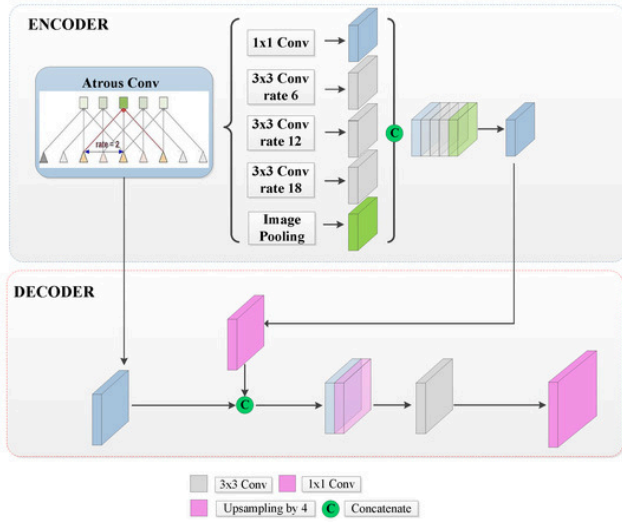


Fig 3. This diagram illustrates the encoder-decoder structure utilised in semantic segmentation, featuring atrous convolutions to capture multi-scale contextual information and a decoder that refines segmentation by concatenating and upsampling features for precise pixel-level output [12].

- **Encoder-Decoder Architecture:** The encoder of DeepLabv3+ employs Atrous Spatial Pyramid Pooling (ASPP) to extract multi-scale contextual information from input images. ASPP applies parallel dilated convolutions at different dilation rates to capture large scales. Employing this method in one layer by itself handles objects of different sizes, a key factor toward correct segmentation in diverse scenes [15]. Similarly, the model increases the field of view by using an atrous convolution that does not downsample the feature map, bypassing typical information loss processes in older pooling methods [15].
- **Application and Relevance:** DeepLabv3+ is very accurate in boundary refinement. It is highly suitable for application in highly complex scenes where several turtles overlap. Such high accuracy in the segmentation of individual turtle parts is a main factor for many downstream analyses related to conservation at large, such as the identification of individuals and behavioural studies.

E. Resnet Model

ResNet is a deep learning architecture introduced by Kaiming He and colleagues in 2015 through their influential paper "Deep Residual Learning for Image Recognition." [27]. This model was designed to address the challenges associated with training very deep neural networks, particularly the vanishing gradient problem and the degradation issue [25], where increasing the depth of a network does not always improve performance. The main innovation in the model was the use of residual, or skip connections. For every block of convolutions, normalizations and/or activation functions, a skip connection was implemented, which aided on the backpropagation of the gradients. The skip connections work by summing the input of the block to the output.

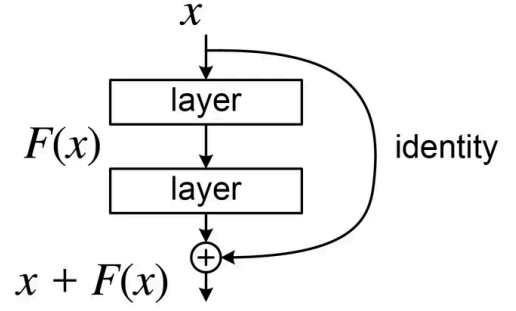


Fig 4. This diagram shows the skip connection of the residual network. It *skips* the block by directly adding a copy of the input to the output.

ResNet is widely used for a variety of tasks in computer vision. It initially demonstrated state-of-the-art results in image classification on benchmarks like ImageNet [30] and has since been adopted for object detection, instance segmentation, and other tasks requiring hierarchical feature extraction. Models like Mask R-CNN and DeepLabv3 often rely on ResNet as their backbone network [31] due to its strong performance in feature representation.

Training ResNet involves using standard optimization techniques like batch normalization and ReLU activations, which stabilize the training process and reduce overfitting. The inclusion of skip connections facilitates the direct flow of gradients to earlier layers, mitigating the risk of vanishing gradients and allowing for efficient training, even in very deep configurations.

F. Related Techniques: Comparative Literature

The below section is concentrated on a comparative analysis of different architectures, devoid of an encoder-decoder structure, with particular attention to YOLO and Mask R-CNN.

With the introduction of The You Only Look Once (YOLO) architecture in 2015, a range of versions have been iterated upon to improve performance [23]. YOLO features a "single-stage approach" extracting both the bounding boxes with associated probability scores in one pass [23][24]. The first version of YOLO features an architecture of "24 convolution layers" with (1 by 1 convolution), "followed by only 2 fully connected layers" [24, pg.9262].

This "simple architecture" which increases speed which in turn propagates a loss in accuracy and efficiency, to the segmentation results. YOLO's main determinant of accuracy is spatially close objects near each other and small objects of interest, where it tests poorly [23][24]. Thus, in the application, YOLO extends itself to tasks that require real-time performance, due to inherent speed.

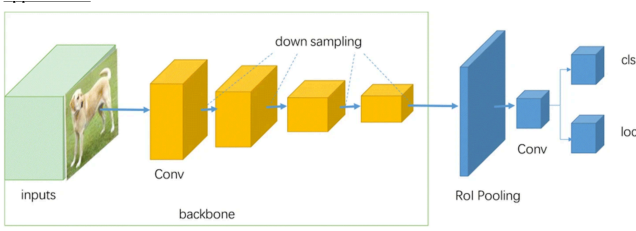


Fig 5. YOLO Architecture diagram [24]

In contrast, Mask region-based fully convolutional networks (Mask R-CNN) is a “two-stage” segmentation model that builds from R-CNN, Fast R-CNN and Faster R-CNN [24]. It was proposed by He et al. [28] to improve region-based convolutional networks by Girshick [29] with a fully convolutional head for mask generation.

The first stage of Faster R-CNN involves the Region Proposal Network (RPN). The RPN generates Region of Interest (RoI) proposals, which are potential bounding boxes around objects. The RPN assigns binary labels to proposals (object vs. background) by evaluating their score against the intersection over union (IOU) [38]. It then refines bounding box coordinates, allowing it to focus only on the areas likely to contain objects.

Faster R-CNN’s second stage takes the proposals from the RPN and applies RoIPool to extract fixed-size feature maps for each RoI (necessary as each region will be of different sizes), which is then used for bounding box regression and classification[41]. Additionally, Mask R-CNN extends this second stage by including a mask to perform instance segmentation [39]. This is done through a small Fully Convolutional Network (FCN) that maintains spatial information, outputting a binary mask for each object [39]. Each of these masks is generated independently (as opposed to multi-class prediction across classes). This design avoids competition among classes, simplifying training and improving segmentation quality.

For instance segmentation, pixel alignment between the input image and RoI features are essential for precise masks. RoIAlign improves on RoIPool, which misaligns pixels due to quantization [39]. The pixel level misalignment provided by the quantisation of ROIpool is mitigated through the use of RoIAlign, which utilises bilinear interpolation, [39]

Every R-CNN model has a big drawback, which each following iteration has tried to mend. That is for every picture, a section of the model must be run for every region proposal, which greatly increases processing time. This is not the case for other segmentation models, as they are not trying to predict the bounding boxes, but only the masks.

G. Conclusion

U-Net and DeepLabv3+ are selected in this work, respectively, based on different strong points of image segmentation provided by the current state of the art. The encoder-decoder architecture is powerful in U-Net for anatomical segmentation with limited data. By incorporating

MobileNetV2-50 into its backbone, U-Net can have much more powerful feature extraction capabilities. Due to the accuracy garnered by U-net, it is frequently used in the medical domain. We propose extending this to the task of turtle segmentation, for precise results.

Meanwhile, DeepLabv3+ provides a robust approach for capturing multi-scale context and refining segmentation boundaries, suitable for the challenges presented by SeaTurtleID2022. They complement each other to provide comprehensive solutions for the accurate segmentation of sea turtles.

III. METHODS

In this project, we have chosen two deep learning-based segmentation approaches for segmenting sea turtles from images, DeepLabv3+ and U-Net. These selections are motivated by the fact that both methods work well on complex images applied to different domains. Now, we briefly give the motivation behind the selection process of each of these two methods and their effectiveness.

A. U-Net

Motivation: U-Net, although originally developed for biomedical image segmentation, the model extends to other fields of application. This is due to its efficiency and precision in segmenting images when the number of samples available for training is limited but highly precise [4]. Its architecture has been designed basically with a symmetric structure using skip connections to preserve spatial hierarchies from high-level to low-level features.

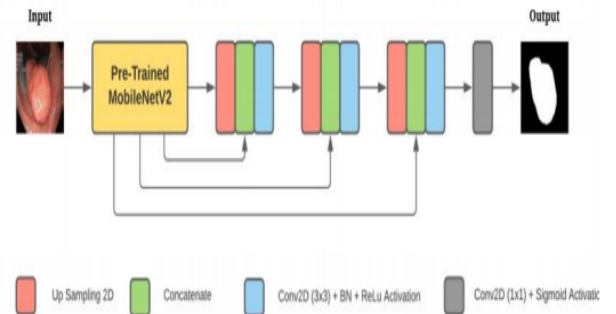


Fig 6. U-Net-Mobile-NetV2 Architecture [34]

MobileNetV2 is optimized for mobile devices, with the main technique involved being depth-wise separable convolutions, which provide a very significant reduction in computational cost and model size without sacrificing too much accuracy [35]. These advanced features include linear bottlenecks and inverted residuals that help preserve the network performance for a wide range of applications, especially on resource-constrained platforms [35].

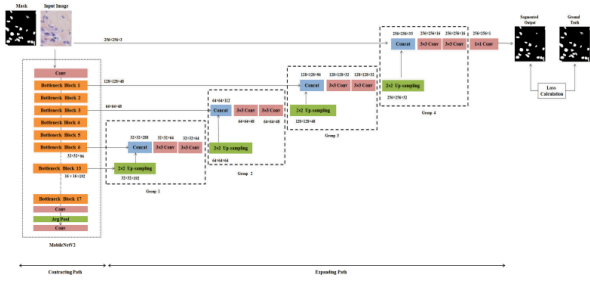


Fig. 7 U-net Model with MobileNetV2 as Encoder [32].

Backbone Network: The decided modification involves using U-Net with MobileNetV2 as the encoder to leverage the efficient means of information processing in MobileNetV2. This combination aims to provide added functionality to U-Net, with a strong and efficient mechanism of feature extraction without sacrificing the depth needed for detailed segmentation tasks such as those required in medical imaging or ecological research [4][35].

Distinctive Features and Achievements: Given the structure, this adoption of MobileNetV2 in the U-Net will operate effectively in complex and fine tasks of imaging, such as sea turtle images where segmentation where catching intricate patterns and textures, is intrinsic to the task. Since MobileNetV2 is lightweight but powerful, this combination will afford it an enhanced ability to process high-resolution images with details without any loss of efficiency and/or accuracy for the system when computational resources are low [35].

Addressing Limitations: Variable conditions, including changing light and partial occlusions, which are commonly observed in natural environments where sea turtles are photographed, pose serious challenges. Efficiency and robustness issues are mitigated with the MobileNetV2, enhancing the feature extraction capability of the U-Net architecture, hence targeting real-time applications and environments where computational power might be limited [33][35].

Relevance: Segmenting the U-Net model with the MobileNetV2 for detailed anatomic studies is highly relevant in the conservation and study of sea turtles. This combination is very beneficial for portable or in-field computing applications where efficiency and accuracy are very much considered paramount needs [4][35].

B. DeepLabv3+

Motivation: DeepLabv3+ is designed to handle complex segmentation tasks by capturing multi-scale contextual information and refining segmentation boundaries. The use of Atrous Spatial Pyramid Pooling (ASPP) enables the model to effectively capture features at different scales, while the decoder helps recover sharp object boundaries.

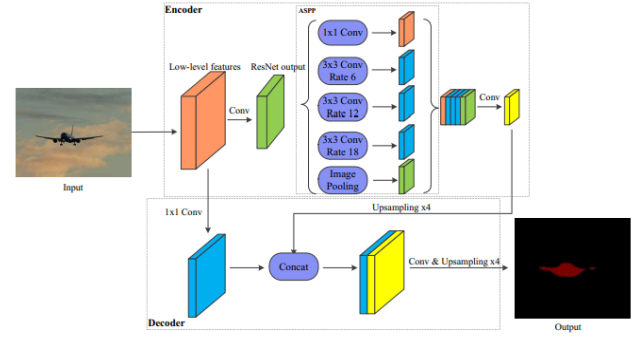


Fig 8. Encoder-Decoder architecture of DeepLab v3+ [40].

Resnet is a lightweight architecture that uses residual connections to improve the backward propagation of the gradients. It avoids vanishing gradients by skipping convolutional layers through the residual connections.

Backbone Network: DeepLabv3+ further modifies the Resnet model to increase the capability of segmentation. Modifications primarily are around depthwise separable convolutions so that spatial features at varying scales could be captured. This personalised approach helps DeepLabv3+ achieve more accurate segmentation results. This proves helpful in scenarios where there are numerous objects with their respective boundaries, for instance, complex scene parsing[14][15].

Distinctive Features and Achievements: The modified DeepLabv3+ with an adapted Resnet backbone uses depth-wise separable convolutions. This is highly efficient in both computations and the number of parameters. This efficiency is needed when dealing with the huge SeaTurtleID2022 dataset, made up of thousands of high-resolution images. This model architecture is particularly optimised to handle multiple scales using atrous convolutions and spatial pyramid pooling, hence being able to adapt to the different sizes and orientations of the turtles captured in the dataset[14][15].

Addressing Limitations: Segmenting flippers, heads, and carapaces requires the differentiation of fine edges and shapes, which usually are blurred or obscured by water. By modifying the Resnet model for the DeepLabv3+, images are treated at multiple scales, taking advantage of contextual information from larger areas in an image to enable the fine segmentation of such details, even under very poor conditions[15].

Relevance: The SeaTurtleID2022 dataset includes diverse natural scenes, often with multiple overlapping turtles. DeepLabv3+ provides precise boundary refinement, allowing accurate segmentation of individual turtle parts, even in challenging environments. This level of precision is crucial for downstream conservation analyses, such as individual identification and behavioural studies.

C. Application in SeaTurtleID2022

The employment of such sophisticated neural network architectures is certain to increase the accuracy and efficiency of segmenting sea turtles from the SeaTurtleID2022 dataset. With this intensive annotation and

gathering over such a long period, this dataset indeed provides an ideal opportunity for these models to learn and generalise complex segmentation tasks across different environmental conditions and turtle behaviours.

D. Conclusion

Segmentation of sea turtles from naturalistic images using MobileNetV2 and the modified Resnet model as the backbones for U-Net and Deeplabv3+ in SeaTurtleID2022 provide a novel approach that addresses the challenges in performing the task.

IV. EXPERIMENTAL RESULTS

Experimental Setup

The segmentation models were evaluated on the SeaTurtleID2022 dataset using open-set splits for training, validation, and testing as defined in the dataset's metadata. This dataset consists of annotated underwater images of turtles, with segmentations provided for three categories: Head, Flippers, and Carapace.

All images were padded and then resized to 256 by 256 pixels for computational efficiency. Throughout the training process, both models were trained using a batch size of 6. The learning rate was set to 0.001. Additionally, the Adam optimiser was used for optimisations.

The backbones of the models were initialised on weights trained on the Imagenet dataset. As the dataset was not big enough for the model to learn the ground truth directly, a fine-tuning approach was required. It necessitated less training time and achieved better results.

It is important to choose the right epoch value. Too few epochs can lead to underfitting, with high training and validation loss due to insufficient learning, while too many epochs can cause overfitting, where the model learns noise, resulting in poor generalisation and higher validation loss.

To figure out the optimal epoch value, we plotted the Loss vs epoch graph. For both U-net and Deeplabv3+, we observed that valid loss consistently decreased until epoch 4 before plateauing as seen below. Training beyond this point did not yield any better performance and increased the risk of overfitting. We can also observe that the loss starts close to the minimum and subsequently is not lowered too much. This is due to initialising the weights on the model trained on the Imagenet dataset.

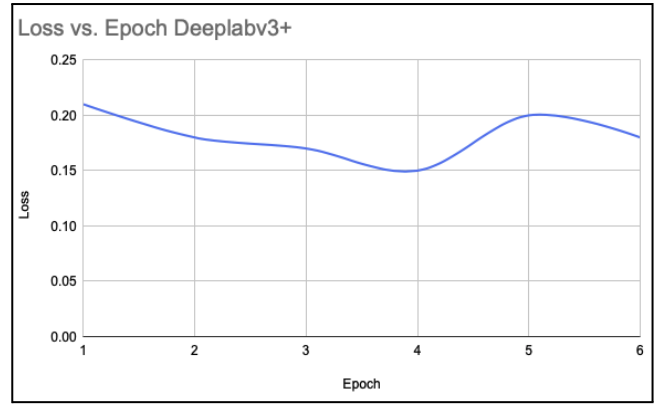


Fig 9. Loss Vs. Epoch for Deeplabv3+

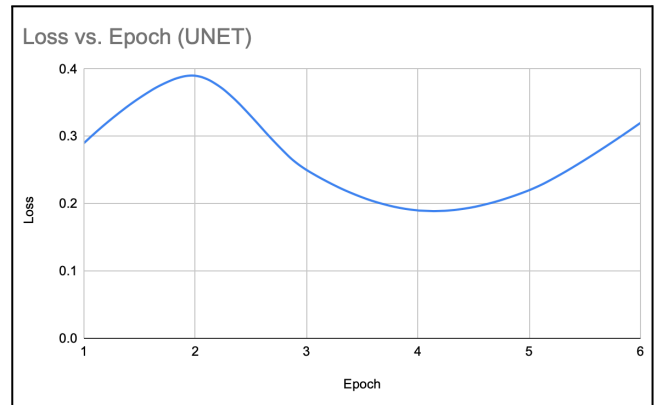


Fig 10. Loss Vs. Epoch for U-Net

For each training image, a multi-class mask was generated using the annotations.json file. The images and their masks were processed into train_data_loader, valid_data_loader, and test_data_loader for the respective dataset splits.

Model Training Considerations

Training efficiency was an important consideration. On the available hardware, U-net required approximately 20 minutes per epoch, while DeepLabV3+ required approximately 30 minutes per epoch due to its more computationally intensive architecture. These training durations highlight the impact of hardware constraints on the project timeline and underscore the importance of high-performance computing resources, such as GPUs, for future iterations.

Model Testing and Evaluation

After training, the models were evaluated on the testing dataset using the test_data_loader. Each test image was fed into the trained models to generate a predicted segmentation mask. These predicted masks were compared against the ground truth masks to calculate the mean Intersection over Union (mIOU) for each category: Head, Flippers, and Carapace.

Quantitative Results

The evaluation results for U-net and DeepLabV3+ are presented in Table 1.

Method	Head mIOU	Flipper mIOU	Carapace mIOU
U-net	0.7252	0.6917	0.8544
DeepLabv3+	0.8237	0.7864	0.8930

Table 1. Result for U-Net and Deeplabv3+

While both models achieved strong segmentation performance, DeepLabV3+ outperformed U-net across all three categories. The superior mIOU scores of DeepLabV3+ may be attributed to its advanced use of atrous convolutions, enabling better capture of the global context. However, U-net demonstrated competitive performance, particularly in segmenting Flippers, where its skip connections likely aided in preserving spatial details.

The higher mIOU for the Carapace reflects the relative simplicity of its shape and contrast with the background. In contrast, Flippers, being more intricate and often occluded, posed greater challenges, resulting in lower mIOU scores for both models.

Qualitative Results

Below are some examples of segmentation masks created by our U-net model and DeepLabV3 model. For comparison, the expected (ground truth) mask is also shown. Above each image, its specific IOU value, for both models, has been listed.

U-net IoUs: head = 0.977; flippers = 0.920; carapace = 0.913 | **DeepLabV3 IoUs:** head = 0.969; flippers = 0.937; Carapace = 0.820

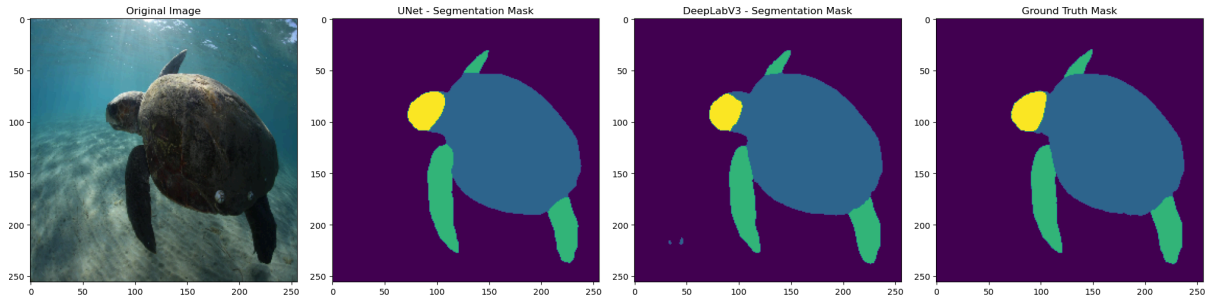


Fig 11. Result 1: Comparison between Original Image, U-Net Segmentation Mask, Deeplabv3+ Segmentation Mask, and Ground Truth Mask

U-net IoUs: head = 0.373; flippers = 0.750; carapace = 0.834 | **DeepLabV3 IoUs:** head = 0.613; flippers = 0.778; Carapace = 0.820

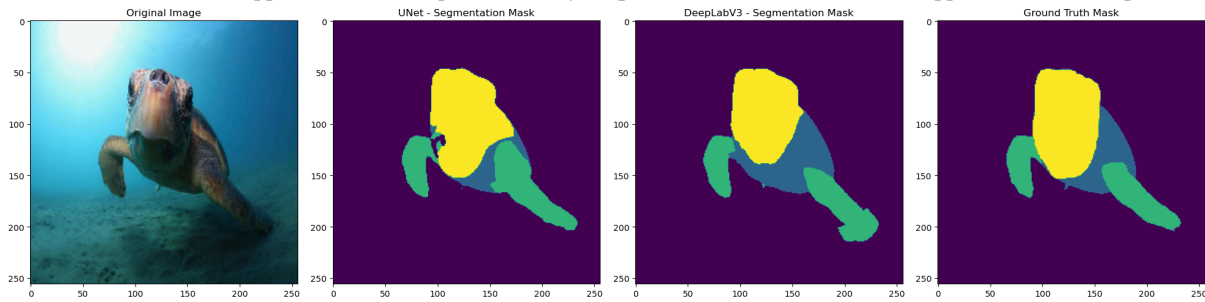


Fig 12. Result 2: Comparison between Original Image, U-Net Segmentation Mask, Deeplabv3+ Segmentation Mask, and Ground Truth Mask

U-net IoUs: head = 0.958; flippers = 0.592; carapace = 0.0 | **DeepLabV3 IoUs:** head = 0.970; flippers = 0.826; Carapace = 0.794

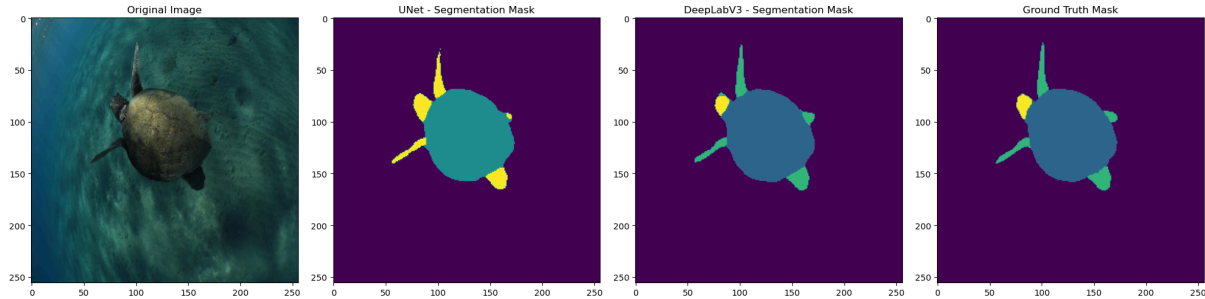


Fig 13. Result 3: Comparison between Original Image, U-Net Segmentation Mask, Deeplabv3+ Segmentation Mask, and Ground Truth Mask

U-net IoUs: head = 0.719; flippers = 0.822; carapace = 0.888 | **DeepLabV3 IoUs:** head = 0.841; flippers = 0.883; Carapace = 0.918

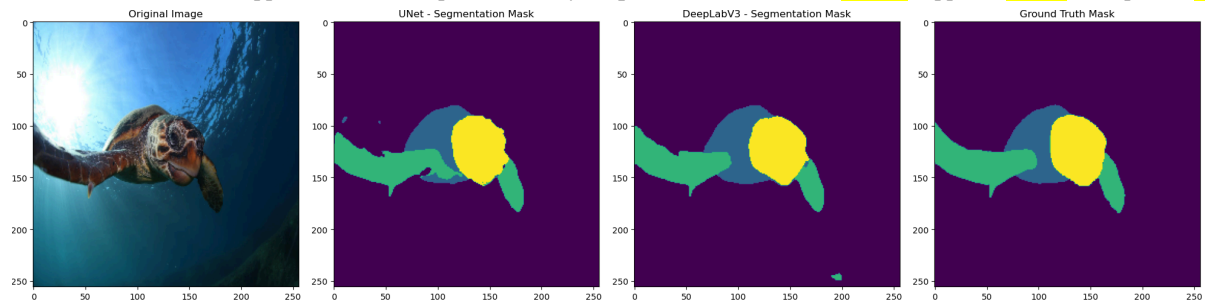


Fig 14. Result 4: Comparison between Original Image, U-Net Segmentation Mask, Deeplabv3+ Segmentation Mask, and Ground Truth Mask

U-net IoUs: head = 0.953; flippers = 0.831; carapace = 0.823 | **DeepLabV3 IoUs:** head = 0.966; flippers = 0.8857; Carapace = 0.831

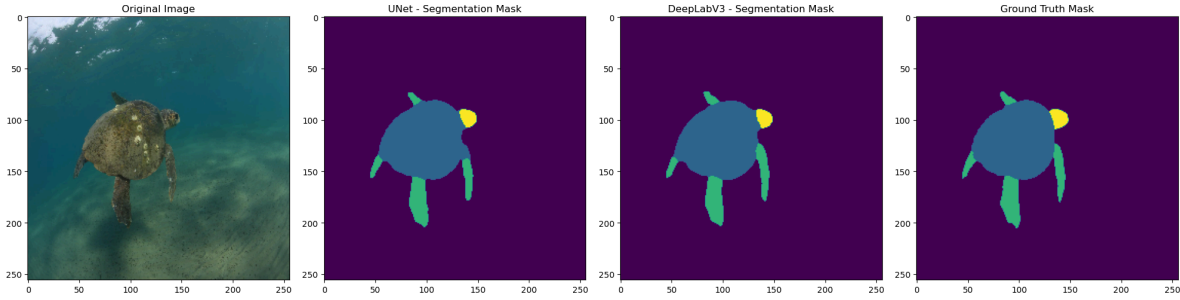


Fig 15. Result 5: Comparison between Original Image, U-Net Segmentation Mask, Deeplabv3+ Segmentation Mask, and Ground Truth Mask

Qualitative analysis revealed that DeepLabV3+ produced more consistent segmentations, particularly for the Carapace, even in images with challenging lighting conditions as seen below. As seen above, U-net struggled with more complex features, such as the Flippers, where intricate boundaries and motion blur posed difficulties. While both models successfully identified the Head and Carapace, U-net occasionally missed finer details, resulting in lower mIOU scores for these categories as opposed to DeepLabV3.

V. DISCUSSION

From Figure 11, poor lighting conditions can be observed, however both models still maintain accurate segmentation. DeepLabV3 avoids segmenting the neck region properly. The second image focal point is the turtle's head which covers the majority of the image. This leads DeepLabV3 to perform better than U-net, with a much closer representation of the ground truth.

For the third image, U-net has misrecognized different categories, The Carapace has been identified as flippers and flippers as the head. This is likely because the third image is uniform in texture, colour and other similar features. Another possible cause of misrecognition is due to a lack of a well-defined edge between the flippers and the carapace.

For the last two images, both models correctly segment the turtle. However, DeepLabV3 outperforms U-net in those too.

U-net demonstrates strong performance in simpler experimentation tasks with clear object boundaries, likely due to its strong focus on local information, which can be seen in examples 1 and 5.

DeepLabV3+'s ability to handle multi-scale data is evident, as can be seen in image 3, in which U-net struggles. In general, it performs better in flippers and heads, which require precise contextual understanding. It also shows the effect of the decoder in polishing the edges of its masks. In all of the examples, but especially in 2 and 4, the masks generated by U-net are rougher and less smooth.

Additionally, in attempts to utilise the mask R CNN architecture, we found slow testing times dismayed our ability to properly test the architecture, forcing us to pivot to DeepLabV3+, due to its compatibility in the data loading process with U-net.

Overall, from the results, we can see that except for the first image, DeepLabV3 has performed better than U-net. However, it is important to note that for many images IoUs do not differ significantly. From these results, it is evident that U-net's performance is equal to or less than DeepLabV3.

VI. CONCLUSION

This project aimed to explore and evaluate the performance of two state-of-the-art segmentation models, U-net and DeepLabV3+, on the SeaTurtleID2022 dataset. While both models demonstrated their capabilities, DeepLabV3+ emerged as the more consistent performer, especially in challenging conditions, such as varying lighting and intricate boundaries. So, DeepLabV3+ outperforms or performs equally well as the U-net method. However, the analysis also highlighted several limitations and areas for improvement that future work can address.

Key determinants and corresponding recommendations include:

Factor 1: A considerable amount of time was concentrated during the model's training process. This was compounded by the limited access to high-performance computing resources like GPU hardware. This led to longer training times, which constrained the time for refining the model's architecture, a crucial aspect of the project.

Recommendation: In the future utilising services like Google Colab which provides more high-performance computing metrics and access to GPU to speed up the process would be ideal to concentrate less time on training, and more time adjusting the architecture of the model.

Factor 2: The models occasionally struggled with small or intricate features, such as flippers, due to the inherent limitations of the loss function used during training. The standard cross-entropy loss focused primarily on overall accuracy, neglecting finer segmentation boundaries.

Recommendation: Implementing alternative loss functions, such as Dice Loss or Focal Loss, in future iterations could improve the models' performance on smaller and more intricate regions. These loss functions emphasise under-represented features and regions with lower confidence, ensuring more accurate segmentation.

Factor 3: The segmentation outputs occasionally exhibited jagged or fragmented edges, particularly for categories like the Flippers. This indicated a lack of refinement in the prediction boundaries.

Recommendation: Employing post-processing techniques such as morphological operations or Conditional Random Fields (CRFs) could help refine segmentation masks by smoothing edges and ensuring contiguous regions. Exploring these techniques in tandem with model predictions would lead to better qualitative results.

By integrating the recommendations proposed, future work can improve the accuracy, generalisability, and practical utility of segmentation models, especially in domains involving challenging real-world datasets like underwater imagery.

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